



RV College of Engineering®
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Question Paper ID: 343768

Course Code: CS373IA

USN

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

VII Semester B.E. Regular Examination February 2026

COMPUTER SCIENCE AND ENGINEERING

Computer Graphics and Virtual Reality

Time : 3 Hours

Instructions to the students

Maximum Marks : 100

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question
No

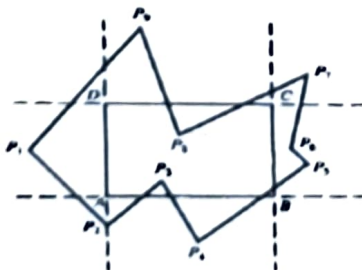
Question

M CO BT

- | | | | | |
|-----|--|----|---|---|
| 1.1 | When to use double buffering? | 01 | 1 | 1 |
| 1.2 | What are the 3 I's of virtual reality? | 01 | 1 | 1 |
| 1.3 | Differentiate between gluOrtho2D() and glOrtho(). | 01 | 1 | 1 |
| 1.4 | Explain what is meant by the term "OpenGL display callback function". | 02 | 2 | 3 |
| 1.5 | Electromagnetic coupling between transmitting antennas and receiving antennas are used to provide information about the _____ of the hand. | 01 | 1 | 1 |
| 1.6 | Mention the drawbacks that were overcome by Bresenham's line over DDA algorithm. | 01 | 1 | 2 |

Apply Sutherland-Hodgeman polygon clipping algorithm for the polygon P1, P2...P8 in Fig (below) against rectangular clipping window. Show the results pictorially.

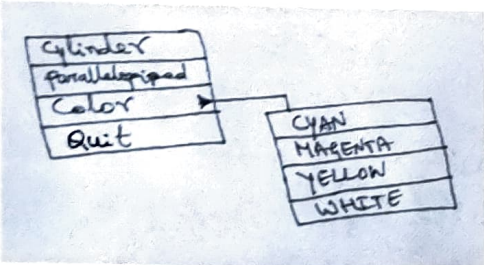
1.7



02 2 2

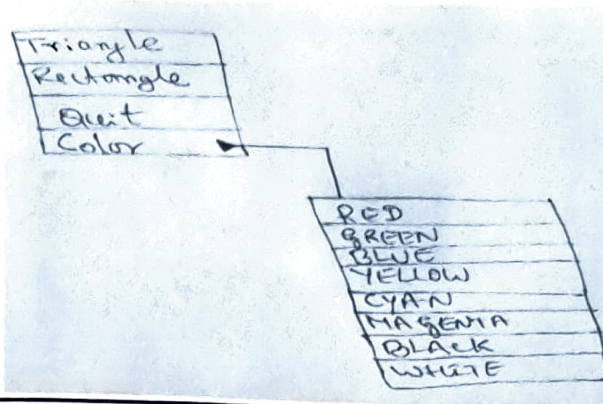
1.8	Show that two or more consecutive rotations can be performed with a single rotation just by adding all the rotation angles.	02	2	3
1.9	A line segment joining the points P1(2, 4) and P2(8, 6). What is the equation of this line in slope intercept form. $y=(1/3)x+(10/3)$	02	2	2
1.10	Define rotation in 3-D transformations.	01	1	1
1.11	Differentiate between parallel projection and perspective projection.	02	1	2
1.12	Describe the basic idea of scan-line method for visible surface detection.	02	1	2
1.13	Distinguish between local and global coordinate systems in object hierarchies.	01	2	2
1.14	What is contact force modeling in physical modelling?	01	2	2

Part B

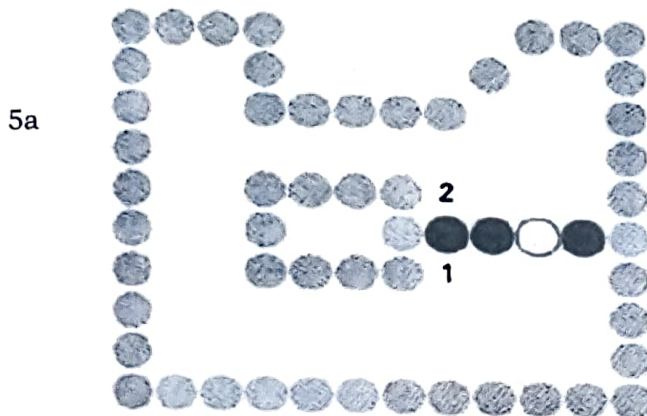
Question No	Question	M	CO	BT
2a	Briefly explain the Display processor architecture with a neat sketch.	04	1	1
2b	All 3D trackers, regardless of the technology they use, share a number of very important performance parameters. Discuss the same illustrating with neat sketches.	04	1	1
2c	What are display lists? How to work with display lists in OpenGL? Give the OpenGL code segment to create a simple car like picture using display lists.	08	3	3
3a	For a line segment, show that clipping against the top of the clipping rectangle can be done independently of the clipping against the other sides. Use this result to show that a clipper can be implemented as a pipeline of four simpler clippers Write the OpenGL program to create the following hierarchical menu. You must give the appropriate code for each menu entries.	06	4	4
3b		10	4	3

OR

4a	How to work with menus and submenus in OpenGL. Explain the following OpenGL API's with their prototype and purpose.	04	1	2
4b	i) glutMouseFunc() ii) glutKeyboardFunc()	04	2	2
4c	Write the OpenGL program to create the following hierarchical menu. You must give the appropriate code for each menu entries.	08	4	3



Examine how pixel spans could be filled using Flood-fill approach (for the 4-connected fill region) for the following polygon. Illustrate the filled pixel spans of the polygon pictorially for 4-connected filled region. Indicate the complete trace of the same using appropriate data structure.



5b Explain the steps involved in line clipping using Liang Barsky algorithm with suitable mathematical formulations.

04 2 3

Reflect the diamond shaped polygon whose vertices are $A(-1,0)$, $B(0,-2)$, $C(1,0)$ and $D(0,2)$ about the horizontal line $y=2$. Determine the new co-ordinates A' , B' , C' and D' on transformation. (Indicate all the steps in computation of resultant transformation).

5c

06 4 4

Also write the OpenGL program segment to perform this reflection (with using Inbuilt APIs for transformations)

OR

6a Write a C program segment to draw a line between the given endpoints using DDA algorithm.

04 3 3

6b Illustrate the working of Sutherland-Hodgman polygon clipping algorithm to clip the given polygon against the clipping window using suitable example.

08 2 3

Devise the procedure to magnify the triangle with vertices $A(0,0)$, $B(1,1)$ and $C(5,2)$ to twice its size while keeping $C(5,2)$ fixed.

6c Implement the function in C (without using OpenGL libraries). Show the expected results of transformation pictorially.

04 3 3

7a	Describe different types of projections used in 3-D graphics.	06	1	2
7b	Describe the scan-line algorithm for visible surface detection.	06	2	2
7c	Derive the rotation matrices for rotation about x, y, and z axes.	04	2	2

OR

8a	The triangle $[ABC] = [(5, 3, 7), (4, 2, 8), (10, 4, 9)]$ is required to be magnified to four times of its original size while keeping $C(10, 4, 9)$ fixed. What are the coordinates of the triangle after magnification.	06	3	3
8b	Explain the steps involved in the 3-D viewing pipeline.	04	1	2
8c	Explain the working of the Z-buffer (depth-buffer) method.	06	2	2

9a	Explain haptic texturing and how it differs when implemented using tactile mice and the PHANTOM device.	06	3	3
9b	Explain how different methods of virtual object shape creation can be applied to develop a realistic virtual environment. Illustrate with suitable examples.	10	3	3

OR

10a	Analyse the role of object hierarchies in managing complex motion and transformations in virtual environments.	08	3	4
10b	In a virtual football game, a player agent must detect opponents, teammates, and the ball to decide whether to dribble, pass, tackle, or shoot. Examine the perception–decision–action cycle in the virtual football player.	08	4	4
